FMO

Horizontal Analysis

Horizontal Analysis –Income Statement

DOVER CORPORATION
Comparative Income Statements
For the Years Ended December 31

			Increase (Decrease)	
	2023	2022	Amount	%
Sales	$ 5,20,000	$ 4,80,000		
Cost of goods sold	3,60,000	3,15,000		
Gross margin	1,60,000	1,65,000		
Operating expenses	1,28,600	1,26,000		
Net operating income	31,400	39,000		
Interest expense	6,400	7,000		
Net income before taxes	25,000	32,000		
Less income taxes (30%)	7,500	9,600		
Net income	$ 17,500	$ 22,400		

Horizontal Analysis

DOVER CORPORATION Comparative Income Statements For the Years Ended December 31				
			Increase (Decrease)	
	2023	2022	Amount	%
Sales	$ 5,20,000	$ 4,80,000	$ 40,000	8.3
Cost of goods sold	3,60,000	3,15,000	45,000	14.3
Gross margin	1,60,000	1,65,000	(5,000)	(3.0)
Operating expenses	1,28,600	1,26,000	2,600	2.1
Net operating income	31,400	39,000	(7,600)	(19.5)
Interest expense	6,400	7,000	(600)	(8.6)
Net income before taxes	25,000	32,000	(7,000)	(21.9)
Less income taxes (30%)	7,500	9,600	(2,100)	(21.9)
Net income	$ 17,500	$ 22,400	$ (4,900)	(21.9)

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Sales increased by 8.3% yet net income decreased by 21.9%.

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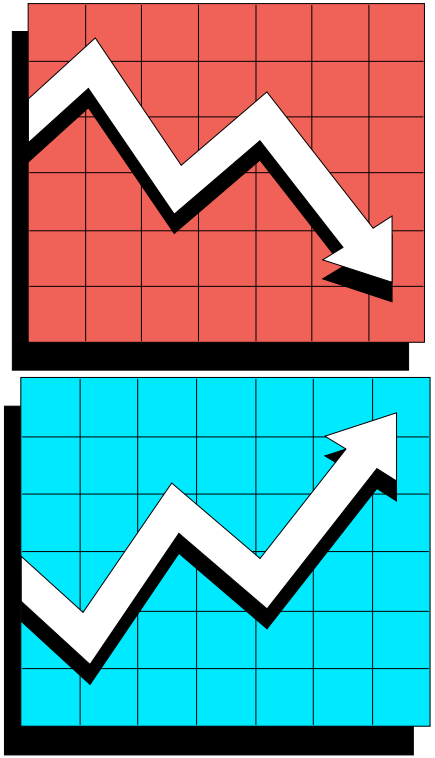
There were increases in both cost of goods sold (14.3%) and operating expenses (2.1%). These increased costs more than offset the increase in sales, yielding an overall decrease in net income.

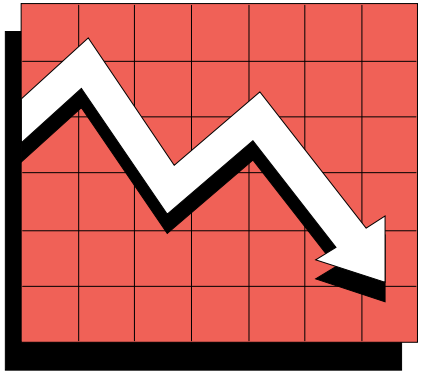
Trend Percentages

Trend percentages state several years' financial data in terms of **a base year, which equals 100 percent.**

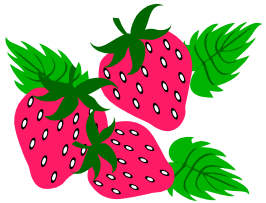


Trend Analysis

$$\text{Trend Percentage} = \frac{\text{Current Year Amount}}{\text{Base Year Amount}} \times 100\%$$



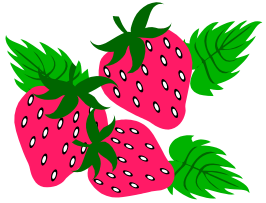
Trend Analysis



Example

Look at the income information for Berry Products for the years 2001 through 2005. We will do a trend analysis on these amounts to see what we can learn about the company.

Trend Analysis



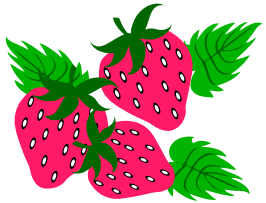
Berry Products
Income Information

For the Years Ended December 31

Item	Year				
	2005	2004	2003	2002	2001
Sales	$ 400,000	$ 355,000	$ 320,000	$ 290,000	$ 275,000
Cost of goods sold	285,000	250,000	225,000	198,000	190,000
Gross margin	115,000	105,000	95,000	92,000	85,000

The base year is 2001, and its amounts will equal 100%.

Trend Analysis



Berry Products
Income Information
For the Years Ended December 31

Item	Year				
	2005	2004	2003	2002	2001
Sales				105%	100%
Cost of goods sold				104%	100%
Gross margin				108%	100%

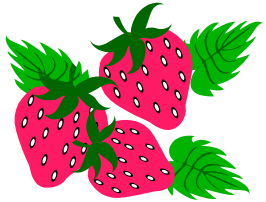
2002 Amount ÷ 2001 Amount × 100%

($290,000 ÷ $275,000) × 100% = 105%

($198,000 ÷ $190,000) × 100% = 104%

($ 92,000 ÷ $ 85,000) × 100% = 108%

Trend Analysis



Berry Products
Income Information

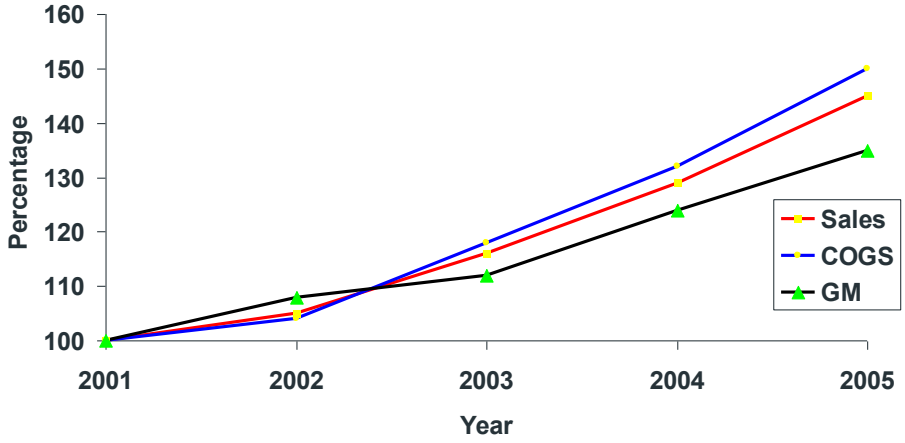
For the Years Ended December 31

Item	Year				
	2005	2004	2003	2002	2001
Sales	145%	129%	116%	105%	100%
Cost of goods sold	150%	132%	118%	104%	100%
Gross margin	135%	124%	112%	108%	100%

By analyzing the trends for Berry Products, we can see that cost of goods sold is increasing faster than sales, which is slowing the increase in gross margin.

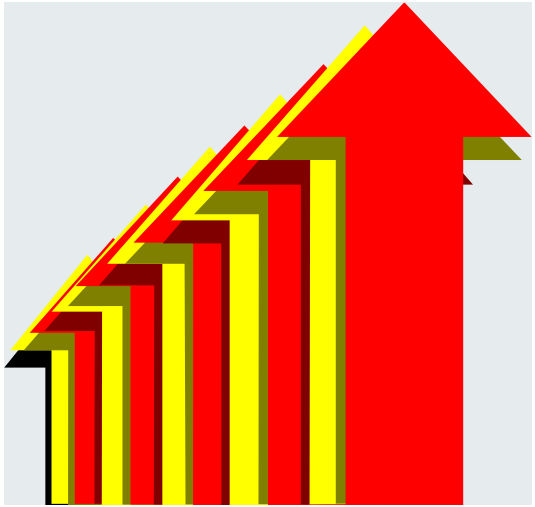
Trend Analysis

We can use the trend percentages to construct a graph so we can see the trend over time.

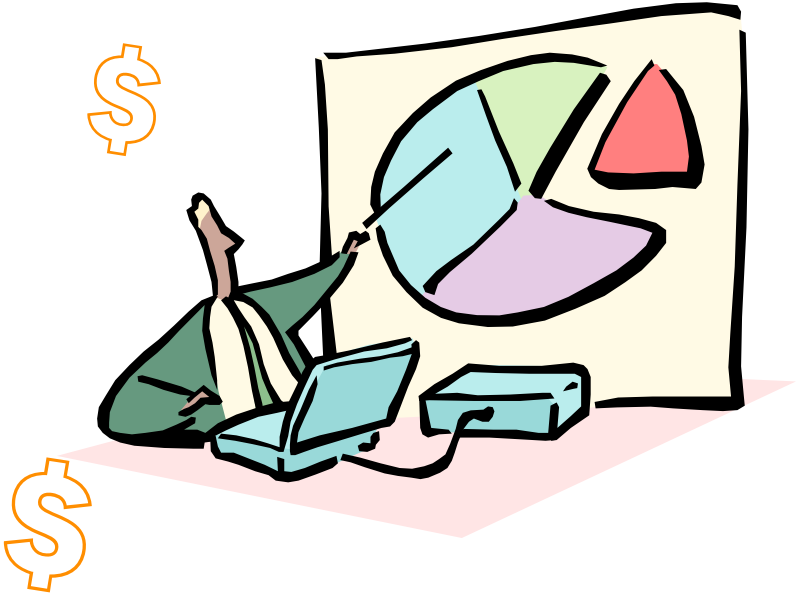


Common-Size Statements

Vertical analysis focuses on the relationships among financial statement items at a given point in time. Common-size financial statement is a vertical analysis in which each financial statement item is expressed as a percentage.



Common-Size Statements



In income statements,

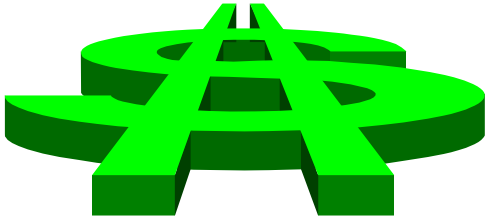
all items are usually expressed

as a percentage of sales.

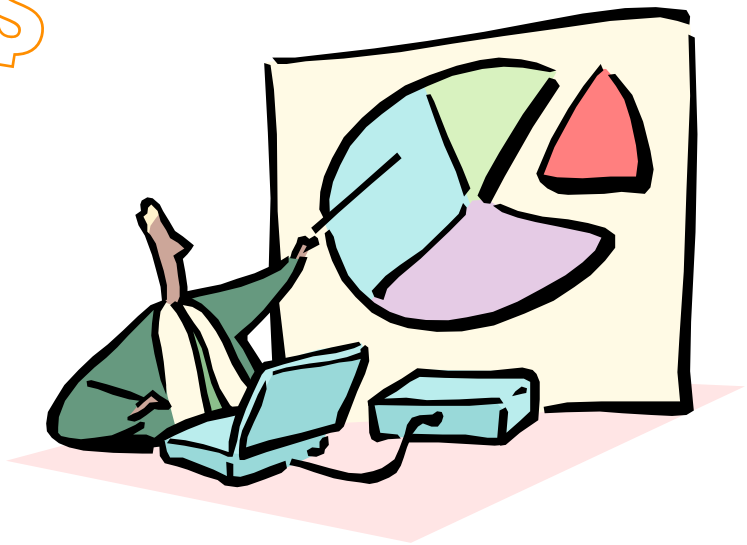
Gross Margin Percentage

$$\text{Gross Margin Percentage} = \frac{\text{Gross Margin}}{\text{Sales}}$$

This measure indicates how much of each sales dollar is left after deducting the cost of goods sold to cover expenses and provide a profit.



Common-Size Statements



In **balance sheets**, all items are usually expressed as a **percentage of total assets**.

Common-Size Statements

	Wendy's		McDonald's	
(dollars in millions)	Dollars	Percentage	Dollars	Percentage
2023 Net income	$ 219	8.00%	$ 893	5.80%

Common-size financial statements are particularly useful when comparing data from different companies.

Common-Size Statements

Example

Let's take another look at the information from the comparative income statements of Clover Corporation for 2005 and 2004.

This time let's prepare common-size statements.

Common-Size Statements –Income Statement

DOVER CORPORATION
Comparative Income Statements
For the Years Ended December 31

			Common-Size Percentages	
	2023	2024	2023	2024
Sales	$ 5,20,000	$ 4,80,000	100.0	100.0
Cost of goods sold	3,60,000	3,15,000		
Gross margin	1,60,000	1,65,000		
Operating expenses	1,28,600	1,26,000		
Net operating income	31,400	39,000		
Interest expense	6,400	7,000		
Net income before taxes	25,000	32,000		
Less income taxes (30%)	7,500	9,600		
Net income	$ 17,500	$ 22,400		

Sales is usually the base and is expressed as 100%.

Common-Size Statements –Income Statement

DOVER CORPORATION
Comparative Income Statements
For the Years Ended December 31

			Common-Size Percentages	
	2023	2024	2023	2024
Sales	$ 5,20,000	$ 4,80,000	100.0	100.0
Cost of goods sold	3,60,000	3,15,000	69.2	65.6
Gross margin	1,60,000	1,65,000		
Operating expenses	1,28,600	1,26,000		
Net				
Inte				
Net				
Less income				
Net income				

2005 Cost ÷ 2005 Sales × 100%
($360,000 ÷ $520,000) × 100% = 69.2%

2004 Cost ÷ 2004 Sales × 100%
($315,000 ÷ $480,000) × 100% = 65.6%

Common-Size Statements –Income Statement

DOVER CORPORATION
Comparative Income Statements
For the Years Ended December 31

What conclusions can we draw?

			Common-Size Percentages	
	2023	2024	2023	2024
Sales	$ 5,20,000	$ 4,80,000	100.0	100.0
Cost of goods sold	3,60,000	3,15,000	69.2	65.6
Gross margin	1,60,000	1,65,000	30.8	34.4
Operating expenses	1,28,600	1,26,000	24.8	26.2
Net operating income	31,400	39,000	6.0	8.2
Interest expense	6,400	7,000	1.2	1.5
Net income before taxes	25,000	32,000	4.8	6.7
Less income taxes (30%)	7,500	9,600	1.4	2.0
Net income	$ 17,500	$ 22,400	3.4	4.7

Quick Check ✓

Which of the following statements describes horizontal analysis?

a. A statement that shows items appearing on it in percentage and dollar form.

b. A side-by-side comparison of two or more years' financial statements.

c. A comparison of the account balances on the current year's financial statements.

d. None of the above.

Quick Check ✓

Horizontal analysis shows the changes between years in the financial data in both dollar and percentage form.

Which of the following statements describes horizontal analysis?

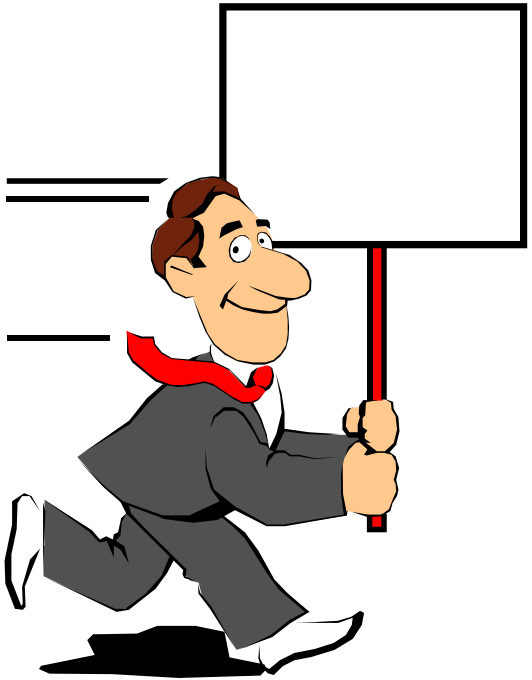
a. A statement that shows items appearing on it in percentage and dollar form.

b. A side-by-side comparison of two or more years' financial statements.

c. A comparison of the account balances on the current year's financial statements.

d. None of the above.

Common-Size Statements –Balance Sheet



Now, let's look at Norton Corporation's 2023 and 2022 financial statements.

Common-Size Statements –Balance Sheet

NORTON CORPORATION Balance Sheets December 31		
	2023	2024
Assets		
Current assets:		
Cash	$ 30,000	$ 20,000
Accounts receivable, net	20,000	17,000
Inventory	12,000	10,000
Prepaid expenses	3,000	2,000
Total current assets	65,000	49,000
Property and equipment:		
Land	1,65,000	1,23,000
Buildings and equipment, net	1,16,390	1,28,000
Total property and equipment	2,81,390	2,51,000
Total assets	$ 3,46,390	$ 3,00,000

Common-Size Statements –Balance Sheet

NORTON CORPORATION Balance Sheets December 31		
	2023	2024
Liabilities and Stockholders' Equity		
Current liabilities:		
Accounts payable	$ 39,000	$ 40,000
Notes payable, short-term	3,000	2,000
Total current liabilities	42,000	42,000
Long-term liabilities:		
Notes payable, long-term	70,000	78,000
Total liabilities	1,12,000	1,20,000
Stockholders' equity:		
Common stock, $1 par value	27,400	17,000
Additional paid-in capital	1,58,100	1,13,000
Total paid-in capital	1,85,500	1,30,000
Retained earnings	48,890	50,000
Total stockholders' equity	2,34,390	1,80,000
Total liabilities and stockholders' equity	$ 3,46,390	$ 3,00,000

Common-Size Statements –Income statement

NORTON CORPORATION
Income Statements
For the Years Ended December 31

	2023	2024
Sales	$ 4,94,000	$ 4,50,000
Cost of goods sold	1,40,000	1,27,000
Gross margin	3,54,000	3,23,000
Operating expenses	2,70,000	2,49,000
Net operating income	84,000	74,000
Interest expense	7,300	8,000
Net income before taxes	76,700	66,000
Less income taxes (30%)	23,010	19,800
Net income	$ 53,690	$ 46,200

Ratio Analysis

Ratio Analysis

- Ratio is a relationship between two or more items of the financial statements. E.g. Net Profit Ratio

- Profit A 65 B 45

- Sales A 650 B 300

Ratio Analysis

A single accounting figure by itself may not communicate any meaningful information

When expressed as a relative to some other figure, it may definitely provide some significant information.

Hence ratio analysis proves very beneficial.

Ratio Analysis

- Ratio analysis is not just comparing different numbers from financial statements.

- It involves comparing the ratio against **previous years**, **against peers**, and with the **industry average** for the purpose of financial analysis.

Ratio Analysis

Ratios help stakeholders (like owners, managers, investors, lenders, employees) to draw conclusion about the

- Performance (past, present and future)
- Strengths & weakness
- And take decision in relation to the firm

Ratio Analysis

1) Liquidity Ratios
2) Return Ratios
3) Capital Structure/Leverage ratios
4) Activity Ratios
5) Efficiency Ratios
6) Profitability Ratios

Ratio Analysis

Liquidity Ratios

Ratio Analysis

- Liquidity ratios are an important class of financial metrics used to determine a debtor's ability to pay off current debt obligations without raising external capital.

- Liquidity ratios determine a company's ability to cover short-term obligations and cash flows,

- Solvency ratios are concerned with a longer-term ability to pay ongoing debt

Current Ratio

1) **Liquidity Ratios:**

i. **Current Ratio = Current Asset/Current Liabilities**

Current Asset (CA)= Inventories + Sundry Debtors + Cash & Bank balances + Receivables / Accruals + Loans & Advances + Disposable Investments

Current Liabilities (CL) = Creditors + Short term Loans + Bank Overdraft + Cash Credit +Outstanding Expenses

Current Ratio

Current Ratio -Does business have enough current assets to meet its current debts ?

Current Ratio

- Current Ratio is a measure of short-term liquidity
- To a creditor, the higher this ratio, the better
- To a firm, a high current ratio may indicate an inefficient use of cash and other short-term assets

Current Ratio

- A generally acceptable current ratio is 1.5:1 or 2:1

- But whether or not a specific ratio is satisfactory depends on the nature of business and characteristics of its CA and CL.

Current Ratio-Case

PRUFROCK CORPORATION 2020 and 2021 Balance Sheets (in millions)	2020	2021	Change
Assets			
Current assets			
Cash	$ 84	$ 146	+$ 62
Accounts receivable	165	188	+ 23
Inventory	393	422	+ 29
Total	$ 642	$ 756	+$114
Fixed assets			
Net plant and equipment	$2,731	$2,880	+$149
Total assets	$3,373	$3,636	+$263
Liabilities and Owners' Equity			
Current liabilities			
Accounts payable	$ 312	$ 344	+$ 32
Notes payable	231	196	− 35
Total	$ 543	$ 540	−$ 3
Long-term debt	$ 531	$ 457	−$ 74
Owners' equity			
Common stock and paid-in surplus	$ 500	$ 550	+$ 50
Retained earnings	1,799	2,089	+ 290
Total	$2,299	$2,639	+$340
Total liabilities and owners' equity	$3,373	$3,636	+$263

PRUFROCK CORPORATION 2021 Income Statement (in millions)		
Sales		$2,311
Cost of goods sold		1,344
Depreciation		276
Earnings before interest and taxes		$ 691
Interest paid		141
Taxable income		$ 550
Taxes (21%)		116
Net income		$ 435
Dividends	$145	
Addition to retained earnings	290	

Current Ratio

2020 Current Ratio: $\frac{642}{543} = 1.18$

2021 Current Ratio: $\frac{756}{540} = 1.40$